Harsh Kumar Mishra

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Education

Sardar Vallabhbhai National Institute of Technology, Surat 2022 – 2027
Integrated Masters Degree, Mathematics *CGPA: 8.06/10*

- **Relevant Coursework:** Probability and Statistics, Linear Algebra, Data Science and Machine Learning

Experience

Gravity Research Pvt. Ltd. March 2025 – June 2025
AI Engineer, Part time *Surat, Gujarat*

`GCP, Pydantic, FastAPI, Pinecone, LangChain, Ollama`

- Engineered a Langchain *ReAct* agent for **Mintzy**, a market analysis tool, with up to **100+** active daily users.
- Built JSON-serializable data models for structured tool call across web-search, prediction and code-execution
- Leveraged namespace isolation and metadata filtering for session-aware context retrieval leading to better responses.

Monday Labs, LLC September 2024 – November 2024
ML Engineer Intern *Remote*

`OpenCV, Tensorflow, PyTesseract, EasyOCR, Colorama`

- **Product 1: DocWizard**
  - Built an automated visual document parsing pipeline to extract colored text and background.
  - Achieved **92%** accuracy in color segmentation, tested across **10+** documents with **500+** rois.
- **Product 2: AadharOCR**
  - Built an automated 4-stage document extraction pipeline to isolate, de-skew, and extract identity fields from ID card scans.
  - Achieved <**500ms** end-to-end processing latency by standardizing inputs to a uniform 600 × 400 geometry, cutting raw pixel volume by >**95%** while maintaining character legibility.

Indian Institute of Technology (ISM), Dhanbad April 2024 – July 2024
Research Intern *Dhanbad, India*

`LangChain, PaddleOCR, ChromaDB, BeautifulSoup, Streamlit`

- Prototyped a RAG pipeline with a robust table extraction module to extract and preserve tabular structure.
- Context aware retrieval with up to **5+** concurrent research papers using semantic search and meta-data filtering.
- Achieved a Hit Rate of **0.85** and NDCG@5 score of **0.82**, tested on **300+** questions over **30+** research papers.

Projects

Modeling of Natural Disaster Economic Losses December 2025

- Harmonized 115+ years of climate (**Berkeley Earth**) and catastrophe records (**EM-DAT**).
- Developed ARIMAX, SARIMAX models with climate exogenous regressors to reduce RMSE by **26%** vs Naive persistence.
- Trained **Random Forest** and **XGBoost** ensembles ($R^2 = 0.828$) and applied **SHAP** (Explainable AI) to decode black-box risk drivers.
- Evaluated models via leak-free **Rolling Forecast Origin** across historical disaster shocks, achieving **60%–75% directional accuracy** and cutting post-shock(2011) forecast error by >**70%**.

EDA: MIMIC-III ICU Patient Data Analysis December 2023

- Cleaned and processed 1,170+ patient records with 51 clinical features from the **MIMIC-III** database.
- Did data visualizations using Seaborn to map survival rates, demographic trends, and the prevalence of 9 key comorbidities.
- Extracted critical clinical insights, identifying an overall ICU mortality rate of over 13% and pinpointing high-risk age groups and diseases.

Achievements

DoTSLASH 8.0 | *National Hackathon, SVNIT* February 2025

- Ranked in the **Top 4% nationwide** for engineering a AI-powered stock market co-pilot.

Skills

Programming: C, C++, Python
ML/DL Libraries: Sci-kit Learn, Tensorflow, PyTorch, OpenCV
Data Analytics: Time Series, Pandas, Seaborn, MatplotLib, SQL
Non Technical: Problem solving, Teamwork, Communication